



Lab Report 6
CSE3023 - Web Application and Development

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Task 1

SampleInsertionRecord.jsp

```
1  <%--
2      Document    : SampleInsertionRecord
3      Created on  : 12 May 2026, 2:05:42 pm
4      Author     : Nazmi
5  --%>
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <%@page language="java"%>
9  <%@page import="java.sql.*"%>
10
11  <!DOCTYPE html>
12  <html>
13  <head>
14      <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
15      <title>JSP Page</title>
16  </head>
17  <body>
18      <h1>Lab 6- Task 1-Sample insertion from Sql to JSP!</h1>
19
20      <%
21          int result;
22          Class.forName("com.mysql.jdbc.Driver");
23          out.println("Step 1: MySQL driver loaded...");
24          %>
25
26      <% String myURL="jdbc:mysql://localhost/CSA3023";
27          Connection myConnection = DriverManager.getConnection(myURL,"root","admin");
28          out.println("Step 2: Database is connected...");
29          %>
30      <br>
31
32      <%
33          out.println("Step 3: prepared Statement created...!");
34          String sInsertQry ="INSERT INTO FirstTable VALUES(?)";
35          PreparedStatement myPS = myConnection.prepareStatement(sInsertQry);
36          %>
37
38      <br>
39      <%
40          out.println("Step 4: Perform insertion of record...!");
41          String name ="Welcome to access MYSQL database with JSP.";
42          myPS.setString(1,name);
43          result = myPS.executeUpdate();
44          if(result > 0){
45              %>
46              <br><!-- comment -->
47              <%
48                  out.println("Step 5: Close database connection...!");
49
50                  out.println(" ");
51                  out.println("Database connection is closed...!");
52
53                  out.print("<p>" + "The record : (" + name
54                      + ") is successfully created..!" + "</p>");
55              }
56              //Step 5: Close database connnection...!
57              myConnection.close();
58          %>
59
60      </body>
61  </html>
```

Output Task 1

Lab 6- Task 1-Sample insertion from Sql to JSP!

Step 1: MySQL driver loaded... Step 2: Database is connected...

Step 3: prepared Statement created...1

Step 4: Perform insertion of record...!

Step 5: Close database connection...! Database connection is closed...!

The record : (Welcome to access MYSQL database with JSP.) is successfully created..!

Task 2

insertAuthor.jsp

```
1 <%--
2 Document : insertAuthor
3 Created on : 12 May 2026, 3:20:49pm
4 Author : Nazmi
5 --%>
6
7 <% page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
8
9 <!DOCTYPE html>
10 <html>
11 <head>
12 <title>Author Registration</title>
13 </head>
14 <body>
15
16 <h1>Lab 6 - Task 2 - Perform creating and retrieving records via JSP page</h1>
17
18 <h2>Author Registration</h2>
19
20 <fieldset>
21 <form action="processAuthor.jsp" method="post">
22
23 Author No (E.g.: UXXXXXX):
24 <input type="text" name="authno"><br><br>
25
26 Name:
27 <input type="text" name="name"><br><br>
28
29 Address:
30 <input type="text" name="address"><br><br>
31
32 City:
33 <input type="text" name="city"><br><br>
34
35 State:
36 <input type="text" name="state"><br><br>
37
38 Zip:
39 <input type="text" name="zip"><br><br>
40
41 <input type="submit" value="Submit">
42 <input type="reset" value="Cancel">
43
44 </form>
45 </fieldset>
46
47 </body>
48 </html>
```

processAuthor.jsp

```
1 <%--
2 Document : processAuthor
3 Created on : 12 May 2026, 3:32:00pm
4 Author : Nazmi
5 --%>
6
7 <%page contentType="text/html" pageEncoding="UTF-8"%>
8 <%page language="java"%>
9 <%page import="java.sql.*"%>
10
11 <jsp:useBean id="myAuthor" class="Lab6.com.Author" scope="request"/>
12 <jsp:setProperty name="myAuthor" property="*" />
13
14 <!DOCTYPE html>
15 <html>
16 <head>
17 <meta charset="UTF-8">
18 <title>Process Author</title>
19 </head>
20 <body>
21
22 <h1>Lab 6 - Task 2 - Perform creating and retrieving records via JSP page</h1>
23
24 <%
25 int result = 0;
26 try {
27
28 Class.forName("com.mysql.jdbc.Driver");
29
30 String myURL = "jdbc:mysql://localhost:3306/CSA3023";
31
32 Connection myConnection =
33 DriverManager.getConnection(myURL, "root", "admin");
34
35 String sInsertQry =
36 "INSERT INTO Author(authno, name, address, city, state, zip) VALUES (?, ?, ?, ?, ?, ?)";
37
38 PreparedStatement myPS =
39 myConnection.prepareStatement(sInsertQry);
40
41 myPS.setString(1, myAuthor.getAuthno());
42 myPS.setString(2, myAuthor.getName());
43 myPS.setString(3, myAuthor.getAddress());
44 myPS.setString(4, myAuthor.getCity());
45 myPS.setString(5, myAuthor.getState());
46 myPS.setString(6, myAuthor.getZip());
47
48 result = myPS.executeUpdate();
49
50 if(result > 0)
51 {
52
53 }
```

```

55 <%
56
57 <h3>Record successfully added into Author table!</h3>
58
59 <p>
60     Record with author no
61     <b><%= myAuthor.getAuthno() %></b>
62     successfully created!
63 </p>
64
65 <p>Name : <%= myAuthor.getName() %></p>
66 <p>Address : <%= myAuthor.getAddress() %></p>
67 <p>City : <%= myAuthor.getCity() %></p>
68 <p>State : <%= myAuthor.getState() %></p>
69 <p>Zip : <%= myAuthor.getZip() %></p>
70
71 <%

```

```

71 <%
72
73
74     myPS.close();
75     myConnection.close();
76
77 }
78 catch(Exception e)
79 {
80     out.println("Error : " + e.getMessage());
81 }
82 <%
83
84 </body>
85 </html>

```

Author.java

```

1  package Lab6.com;
2
3  public class Author {
4
5      private String authno;
6      private String name;
7      private String address;
8      private String city;
9      private String state;
10     private String zip;
11
12     public Author() {
13     }
14
15     public String getAuthno() {
16         return authno;
17     }
18
19     public void setAuthno(String authno) {
20         this.authno = authno;
21     }
22
23     public String getName() {
24         return name;
25     }
26
27     public void setName(String name) {
28         this.name = name;
29     }

```

```

31     public String getAddress() {
32         return address;
33     }
34
35     public void setAddress(String address) {
36         this.address = address;
37     }
38
39     public String getCity() {
40         return city;
41     }
42
43     public void setCity(String city) {
44         this.city = city;
45     }
46
47     public String getState() {
48         return state;
49     }
50
51     public void setState(String state) {
52         this.state = state;
53     }
54
55     public String getZip() {
56         return zip;
57     }
58
59     public void setZip(String zip) {
60         this.zip = zip;
61     }
62 }

```

Output Task 2:

Lab 6 - Task 2 - Perform creating and retrieving records via JSP page

Author Registration

Author No (E.g.: UKXXXXX):	Iman
Name:	Aiman
Address:	Lot depan rumah
City:	Kuala nerus
State:	Ganu
Zip:	16010
Submit Cancel	

Lab 6 - Task 2 - Perform creating and retrieving records via JSP page

Record successfully added into Author table!

Record with author no **Iman** successfully created!

Name : Aiman

Address : Lot depan rumah

City : Kuala nerus

State : Ganu

Zip : 16010

Task 3

insertStudent.jsp

```
1 <!--
2     Document    : InsertStudent
3     Created on  : 12 May 2026, 4:01:19 pm
4     Author     : Nazmi
5 -->
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11     <meta charset="UTF-8">
12     <title>Student Registration</title>
13
14     <style>
15         body{
16             font-family: Arial;
17         }
18
19         fieldset{
20             width: 500px;
21         }
22
23         table{
24             padding: 10px;
25         }
26
27         td{
28             padding: 8px;
29         }
30     </style>
31
32 </head>
```

```
34 <body>
35
36     <h2>Insert Student </h2>
37
38     <fieldset>
39
40         <form action="processStudent.jsp" method="post">
41
42             <table>
43
44                 <tr>
45                     <td>Student No</td>
46
47                     <td>
48                         <input type="text"
49                             name="studentNo"
50                             placeholder="E.g.: UK000XX"
51                             pattern="[A-Z0-9]*"
52                             required>
53                     </td>
54                 </tr>
55
56                 <tr>
57                     <td>Name</td>
58
59                     <td>
60                         <input type="text"
61                             name="studentName"
62                             placeholder="Enter your name"
63                             required>
64                     </td>
65                 </tr>
```

```
67
68         <td>Program</td>
69
70         <td>
71             <select name="program">
72
73                 <option>BSc. Soft. Eng.</option>
74                 <option>BSc. with IM</option>
75                 <option>BSc. in Networking</option>
76                 <option>BSc. in Robotics</option>
77             </select>
78         </td>
79     </tr>
80
81     <tr>
82         <td colspan="2">
83
84             <input type="submit" value="Submit">
85
86             <input type="reset" value="Cancel">
87
88         </td>
89     </tr>
90
91 </table>
92
93 </form>
94
95 </fieldset>
96
97 </body>
```

processStudent.jsp

```
1  Document : processStudent
2  Created on : 12 May 2026, 9:49:20 pm
3  Author : Nazmi
4  -->
5
6  <%page contentType="text/html" pageEncoding="UTF-8"%>
7  <%page language="java"%>
8  <%page import="java.sql.*"%>
9  <%page errorPage="errorStudent.jsp"%>
10
11  <!-- Create an object for Student -->
12  <jsp:useBean id="myStudent" class="Lab6.com.Student" scope="request"/>
13
14  <!-- Assign data entry from page insertStudent.jsp page into Student's bean -->
15  <jsp:setProperty name="myStudent" property="*"/>
16
17  <%
18      int result;
19
20      try {
21          //Step 1: Load JDBC driver...
22          Class.forName("com.mysql.cj.jdbc.Driver");
23          System.out.println("Step 1: MySQL driver loaded...!");
24
25          //Step 2: Establish the connection...
26          String myURL = "jdbc:mysql://localhost:3306/CSA3023";
27          Connection myConnection = DriverManager.getConnection(myURL, "root", "admin");
28          System.out.println("Step 2: Database is connected...!");
29
30      } catch (Exception e) {
31          out.println(e);
32      }
33  %>
```

```
34  //Step 3: Create a PreparedStatement object...
35  System.out.println("Step 3: Prepared Statements created...!");
36
37  //Prepared SQL Query as a String...
38  String sInsertQry = "INSERT INTO Student(stuno, stuname, stuprogram) VALUES(?, ?, ?)";
39  System.out.println("\tSQL Query: " + sInsertQry);
40
41  //Call method preparedStatement
42  PreparedStatement myPS = myConnection.prepareStatement(sInsertQry);
43
44  //Assign each value to respective columns for Student's table...
45  System.out.println("Step 4: Perform insertion of record...!");
46  myPS.setString(1, myStudent.getStuno());
47  myPS.setString(2, myStudent.getName());
48  myPS.setString(3, myStudent.getProgram());
49
50  //Execute query
51  result = myPS.executeUpdate();
52
53  if ( result > 0 )
54  {
55      System.out.println("\tRecord successfully added into Student's table...!");
56      out.print("<p> " + "Record with student no " + myStudent.getStuno() +
57              " successfully created..!" + "</p>");
58      out.print("<p> " + "Details of record are: " + "</p>");
59      out.print("<p>Student ID : " + myStudent.getStuno() + "</p>");
60      out.print("<p>Name : " + myStudent.getName() + "</p>");
61      out.print("<p>Program : " + myStudent.getProgram() + "</p>");
62  }
```

```
63  //Close database connection
64  System.out.println("Step 5: Close database connection...!");
65  myConnection.close();
66  System.out.println(" ");
67  System.out.println("Database connection is closed...!");
68
69  } catch (Exception e) {
70      out.println(e);
71  }
72  %>
```

student.java

```
1  package Lab6.com;
2
3  import java.util.regex.*;
4
5  public class Student {
6
7      private String stuno;
8      private String name;
9      private String program;
10
11      public Student() {
12      }
13
14      public String getStuno() {
15
16          Pattern pt = Pattern.compile("[A-Z0-9]*");
17          Matcher mt = pt.matcher(this.stuno);
18
19          boolean bl = mt.matches();
20
21          if(bl){
22              return stuno;
23          }
24          else{
25              return "Invalid Format";
26          }
27      }
28
29      public void setStuno(String stuno) {
30          this.stuno = stuno;
31      }
```

```
33      public String getName() {
34          return name;
35      }
36
37      public void setName(String name) {
38          this.name = name;
39      }
40
41      public String getProgram() {
42          return program;
43      }
44
45      public void setProgram(String program) {
46          this.program = program;
47      }
48  }
```

errorStudent.jsp

```
1  <!--
2      Document      : errorStudent
3      Created on    : 12 May 2026, 9:55:13 pm
4      Author       : Nazmi
5  -->
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <!DOCTYPE html>
9  <html>
10     <head>
11         <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12         <title>JSP Page</title>
13     </head>
14     <body>
15         <form id="errorFrm" action="insertStudent.jsp" method="post">
16             <h1>Lab 9 - Task 1 - Perform creating and retrieving records via JSP page</h1>
17             <p>Error occurred when inserting record...!</p>
18             <p><jsp:expression> exception.getMessage() </jsp:expression></p>
19             <br>
20         </form>
21     </body>
22 </html>
```

Task 3 Output:

Insert Student

Student No	S121212
Name	yum
Program	BSc. Soft. Eng. ▼
Submit Cancel	

Record with student no S121212 successfully created..!

Details of record are;

Student ID : S121212

Name : yum

Program : BSc. Soft. Eng.

Task 4

queryStudent.jsp

```

8  <%@page contentType="text/html" pageEncoding="UTF-8"%>
9  <%@page import="java.sql.*"%>
10
11  <!DOCTYPE html>
12  <html>
13  <head>
14      <meta charset="UTF-8">
15      <title>Student Records</title>
16
17      <style>
18          table {
19              border-collapse: collapse;
20          }
21
22          td, th {
23              border: 1px solid #999;
24              padding: 0.5rem;
25              text-align: left;
26          }
27
28          th {
29              background: gold;
30          }
31      </style>
32  </head>
33  <body>
34
35      // Use a Java scriptlet to create a simple structure of HTML table.
36      out.print("<table>");
37      out.print("<thead>");
38      out.print("<tr>");
39      out.print("<th>" + "ISBNNo" + "</th>");
40      out.print("<th>" + "Author" + "</th>");
41      out.print("<th>" + "Title" + "</th>");
42      out.print("</tr>");
43      out.print("</thead>");
44      out.print("<tbody>");
45
46      try {
47          // Step 1: Load JDBC driver...
48          Class.forName("com.mysql.jdbc.Driver");
49          System.out.println("Step 1: MySQL driver loaded...!");
50
51          // Step 2: Establish the connection...
52          String myURL = "jdbc:mysql://localhost/CSA3023";
53          Connection myConnection = DriverManager.getConnection(myURL, "root", "admin");
54          System.out.println("Step 2: Database is connected...!");
55
56          // Step 3: Create a statement object...
57          Statement myStatement = myConnection.createStatement();
58
59          // Step 4: Perform retrieve record from Student's table... (R-Retrieve)
60          String myQuery = "SELECT * FROM student";
61          ResultSet myResultSet = myStatement.executeQuery(myQuery);
62
63          // Fetch the record into HTML's table.
64          while ( myResultSet.next() )
65          {
66              out.print("<tr>");
67              out.print("<td width='20%'>" + myResultSet.getString(1) + "</td>");
68              out.print("<td width='40%'>" + myResultSet.getString(2) + "</td>");
69              out.print("<td width='40%'>" + myResultSet.getString(3) + "</td>");
70              out.print("</tr>");
71          }
72
73          // Step 5: Close database connection...!
74          System.out.println("Step 5: Close database connection...!");
75          myConnection.close();
76          System.out.println(" ");
77          System.out.println("Database connection is closed...!");
78
79          } catch (Exception e) {
80              e.printStackTrace();
81          }
82
83          out.print("</tbody>");
84          out.print("</table>");
85      }
86  </body>
87  </html>

```

Output task 4

ISBNNo	Author	Title
S121212	yum	BSc. Soft. Eng.
S75948	Nazmi	BSc. Soft. Eng.
UK12489	Ahmad Salam	BSc. with IM
UK56789	Rosnah Azman	BSc. Soft. Eng.
UK67342	Liew Cheng Huat	BSc. in Robotics
UK88888	Mohamad Nor	BSc. in Networking

Task 5

registerMarathon.jsp

```
1 <%--
2 Document : registerMarathon
3 Created on : 12 May 2026, 10:44:51 pm
4 Author : Nazmi
5 --%>
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12 <title>JSP Page</title>
13 </head>
14 <body>
15
16 <form action="processMarathon.jsp" method="post">
17 <fieldset style="width: 50%;">
18 <legend>Marathon Registration</legend>
19 <table>
20 <tr>
21 <td>IC No</td>
22 <td><input type="text" name="icno" placeholder="E.g.: 921110-10-2514"></td>
23 </tr>
24 <tr>
25 <td>Name</td>
26 <td><input type="text" name="name" placeholder="Enter your name"></td>
27 </tr>
28 </table>
```

```
29 <td>Category</td>
30 <td>
31 <select name="category">
32 <option value="5 KM">5 KM</option>
33 <option value="7 KM">7 KM</option>
34 <option value="10 KM">10 KM</option>
35 </select>
36 </td>
37 </tr>
38 <tr>
39 <td colspan="2">
40 <input type="submit" value="Submit">
41 <input type="reset" value="Cancel">
42 </td>
43 </tr>
44 </table>
45 </fieldset>
46 </form>
47 </body>
48 </html>
49
```

processMarathon.jsp

```
1 <%--
2 Document : processMarathon
3 Created on : 12 May 2026, 11:17:24 pm
4 Author : Nazmi
5 --%>
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <%@page import="java.sql.*"%>
9 <%@page import="Lab6.com.Database"%>
10 <%@page import="Lab6.com.Marathon"%>
11 <%@page import="Lab6.com.MarathonDAO"%>
12
13 <jsp:useBean id="myMarathon" class="Lab6.com.Marathon" scope="request"/>
14 <jsp:setProperty name="myMarathon" property="*"/>
```

```
15 <%
16 int result;
17
18 // Step 1: Create Database object...
19 Database myDB = new Database();
20 MarathonDAO object1 = new MarathonDAO();
21
22 // Step 2: Add the records...
23 result = object1.addDetails(myMarathon);
24
25 // Step 3: Determine whether the transaction is success...
26 if (result > 0) {
27 System.out.println("\tRecord successfully added into Book's table...!");
28 out.print("<p>" + "Record with IC No " + myMarathon.getIcno() +
29 " successfully created..!" + "</p>");
30 out.print("<p>" + "Details of record are: " + "</p>");
31 out.print("<p>IC No : " + myMarathon.getIcno() + "</p>");
32 out.print("<p>Name : " + myMarathon.getName() + "</p>");
33 out.print("<p>Category : " + myMarathon.getCategory() + "</p>");
34 }
35
36 // Step 4: Close database connection...!
37 System.out.println("Step 5: Close database connection...!");
38 myDB.closeConnection();
39 System.out.println(" ");
40 System.out.println("Database connection is closed...!");
41
42 %>
```

Marathon.java

```
2 package Lab6.com;
3
4 /**
5  *
6  * @author Nazmi
7  */
8 public class Marathon {
9
10     private String icno;
11     private String name;
12     private String category;
13
14     public Marathon() {
15     }
16
17     public String getIcno() {
18         return icno;
19     }
20
21     public void setIcno(String icno) {
22         this.icno = icno;
23     }
24
25     public String getName() {
26         return name;
27     }
28
29     public void setName(String name) {
30         this.name = name;
31     }
```

```
33     public String getCategory() {
34         return category;
35     }
36
37     public void setCategory(String category) {
38         this.category = category;
39     }
40 }
```

marathonDAO.java

```
5 package Lab6.com;
6
7 import java.sql.Connection;
8 import java.sql.PreparedStatement;
9 import java.sql.SQLException;
10 import Lab6.com.Database;
11
12 public class MarathonDAO {
13     private Connection connection;
14     private int result = 0;
15
16     public MarathonDAO() throws ClassNotFoundException {
17         connection = Database.getConnection();
18     }
19
20     public int addDetails(Marathon marathon) {
21         try {
22             String mySQL = "INSERT INTO marathon(icno, name, category) values (?, ?, ?)";
23             PreparedStatement preparedStatement = connection.prepareStatement(mySQL);
24
25             System.out.println("IC No    = " + marathon.getIcno());
26             System.out.println("Name    = " + marathon.getName());
27             System.out.println("Category = " + marathon.getCategory());
28
29             // Parameters
30             preparedStatement.setString(1, marathon.getIcno());
31             preparedStatement.setString(2, marathon.getName());
32             preparedStatement.setString(3, marathon.getCategory());
33
34             result = preparedStatement.executeUpdate();
35
36         } catch (SQLException e) {
37             e.printStackTrace();
38         }
39         return result;
40     }
41 }
```

Database.java

```
11 import java.sql.Connection;
12 import java.sql.DriverManager;
13 import java.sql.PreparedStatement;
14 import java.sql.SQLException;
15 import java.util.logging.Level;
16 import java.util.logging.Logger;
17 import Lab6.com.Marathon;
18
19 public class Database {
20     private static Connection myConnection = null;
21     private static String myURL = "jdbc:mysql://localhost:3306/CSA3023";
22     private int result = 0;
23
24     public static Connection getConnection() throws ClassNotFoundException {
25         if (myConnection != null) {
26             return myConnection;
27         } else {
28             try {
29                 Class.forName("com.mysql.jdbc.Driver");
30                 myConnection = DriverManager.getConnection(myURL, "root", "a");
31             } catch (SQLException e) {
32                 e.printStackTrace();
33             }
34         }
35         return myConnection;
36     }
37
38     public void closeConnection() throws ClassNotFoundException {
39         try {
40             myConnection.close();
41         } catch (SQLException e) {
42             e.printStackTrace();
43         }
44     }
45 }
```

Output Task 5

localhost:8080/lab6/registerMarathon.jsp

Marathon Registration

IC No

Name

Category

← ↻ ⓘ localhost:8080/lab6/processMarathon.jsp

Record with IC No 05020303102321 successfully created..!

Details of record are;

Ic No : 05020303102321

Name : Aiman

Category : 5 KM

LabExercise

InsertUser.html

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>User Registration</title>
5  </head>
6  <body>
7
8      <h2>User Registration Form</h2>
9
10     <form action="processUser.jsp" method="post">
11
12         Username :
13         <input type="text" name="username" required>
14         <br><br>
15
16         Password :
17         <input type="password" name="password" required>
18         <br><br>
19
20         First Name :
21         <input type="text" name="firstname" required>
22         <br><br>
23         Last Name :
24         <input type="text" name="lastname" required>
25         <br><br>
26
27         <input type="submit" value="Register">
28
29     </form>
30
31 </body>
32 </html>
```

Login.jsp

```
1  <!--
2      Document    : login
3      Created on   : 12 May 2026, 9:38:57 pm
4      Author      : Nazmi
5  -->
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <!DOCTYPE html>
9  <html>
10 <head>
11     <title>User Login</title>
12 </head>
13 <body>
14
15     <h2>Login Form</h2>
16
17     <form action="doLogin.jsp" method="post">
18
19         Username :
20         <input type="text" name="username" required>
21         <br><br>
22
23         Password :
24         <input type="password" name="password" required>
25         <br><br>
26
27         <input type="submit" value="Login">
28
29     </form>
30
31 </body>
32 </html>
```

Main.jsp

```
1  <!--
2      Document    : main
3      Created on  : 12 May 2026, 9:41:44 pm
4      Author     : Nazmi
5  -->
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8
9  <!DOCTYPE html>
10 <html>
11 <head>
12     <title>Main Page</title>
13 </head>
14 <body>
15
16     <h2>Welcome to the System</h2>
17
18     Username :
19     <%= session.getAttribute("username") %>
20
21     <br><br>
22
23     First Name :
24     <%= session.getAttribute("firstname") %>
25
26     <br><br>
27
28     Last Name :
29     <%= session.getAttribute("lastname") %>
30
31 </body>
32 </html>
```

processUser.jsp

```
1  <!--
2      Document   : processUser
3      Created on : 12 May 2026, 9:39:56 pm
4      Author    : Nazmi
5  -->
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <%@page import="java.sql.*"%>
9
10 <!DOCTYPE html>
11 <html>
12 <head>
13     <title>Process User</title>
14 </head>
15 <body>
16
17 <%
18     String username = request.getParameter("username");
19     String password = request.getParameter("password");
20     String firstname = request.getParameter("firstname");
21     String lastname = request.getParameter("lastname");
22
23     Connection conn = null;
24     PreparedStatement ps = null;
```

```
25
26     try {
27
28         Class.forName("com.mysql.jdbc.Driver");
29
30         conn = DriverManager.getConnection(
31             "jdbc:mysql://localhost:3306/CSA3023",
32             "root",
33             "admin"
34         );
35
36         String sql = "INSERT INTO userprofile VALUES (?, ?, ?, ?)";
37
38         ps = conn.prepareStatement(sql);
39
40         ps.setString(1, username);
41         ps.setString(2, password);
42         ps.setString(3, firstname);
43         ps.setString(4, lastname);
44
45         int result = ps.executeUpdate();
46
47         if(result > 0){
48             out.println("<h3>User registered successfully!</h3>");
49             out.println("<a href='login.jsp'>Login Here</a>");
50         }
51         else{
52             out.println("Registration failed.");
53         }
54     }
55     catch(Exception e){
56         out.println("Error : " + e.getMessage());
57     }
58 }
```

```
59     finally{
60
61         if(ps != null)
62             ps.close();
63
64         if(conn != null)
65             conn.close();
66     }
67 <%>
68
69 </body>
70 </html>
```


Error.jsp

```
1  <%--
2      Document    : error
3      Created on  : 12 May 2026, 9:42:10?pm
4      Author     : Nazmi
5  --%>
6
7  <!DOCTYPE html>
8  <html>
9  <head>
10     <title>Login Failed</title>
11 </head>
12 <body>
13
14     <h2>Invalid Username or Password</h2>
15
16     <a href="login.jsp">Try Again</a>
17
18 </body>
19 </html>
```

doLogin.jsp

```
1  Document : doLogin
2  Created on : 12 May 2026, 9:41:08 pm
3  Author : Nazmi
4
5  <!--
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <%@page import="java.sql.*"%>
9
10 <%
11     String username = request.getParameter("username");
12     String password = request.getParameter("password");
13
14     Connection conn = null;
15     PreparedStatement ps = null;
16     ResultSet rs = null;
17
18     try{
19
20         Class.forName("com.mysql.jdbc.Driver");
21
22         conn = DriverManager.getConnection(
23             "jdbc:mysql://localhost:3306/CSA3023",
24             "root",
25             "admin"
26         );
27
28         String sql = "SELECT * FROM userprofile "
29             + "WHERE username=? AND password=?";
30
31         ps = conn.prepareStatement(sql);
32
33         ps.setString(1, username);
34         ps.setString(2, password);
35
36         rs = ps.executeQuery();
37
38         if(rs.next()){
39
40             session.setAttribute("username",
41                 rs.getString("username"));
42
43             session.setAttribute("firstname",
44                 rs.getString("firstname"));
45
46             session.setAttribute("lastname",
47                 rs.getString("lastname"));
48
49             response.sendRedirect("main.jsp");
50         }
51         else{
52
53             response.sendRedirect("error.jsp");
54         }
55     }
56     catch(Exception e){
57
58         out.println("Error : " + e.getMessage());
59     }
60 }
61 %>
```

Output:

User Registration Form

Username :

Password :

First Name :

Last Name :

User registered successfully!

[Login Here](#)

Welcome to the System

Username : Nazmi12

First Name : Nazmi12

Last Name : Chan12

Invalid Username or Password

[Try Again](#)